

Classwork

Dataset: palmerpenguins (built into Seaborn)

Task 1 - Load & Explore

1. Load the dataset

```
import seaborn as sns
# Load the penguins dataset into a pandas DataFrame
penguins = sns.load_dataset("penguins")
```

2. See the basic description of the dataset (shape, info, describe)

Task 2 - Data Quality Checks

1. Check for missing values
2. Check unique values in the categorical columns, and do a groupby operation to explore the data
3. Check for duplicate rows

Task 3 - Exploratory Data Analysis

1. Visualize the target (species) distribution - is it balanced?
2. Plot histograms or boxplots of the numeric features
3. Create a pairplot colored by species

Task 4 - Preprocessing

1. Handle the missing values you found in Task 2 (justify your approach)
2. Encode the categorical feature(s)
3. Compute a correlation matrix - which features look most useful?

Submission:

1. Screenshot of your puffer platform
2. Pdf file of the solution